

MICHAEL WIECK-SOSA

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EDUCATION

Carnegie Mellon University | PhD in Statistics | Advisor: [Aaditya Ramdas](#) May 2027

- GPA: 3.98/4.00 | Awards: DeGroot-Goel Fellowship (2026), awarded by faculty to one Statistics PhD student per year
- Thesis topics: prediction with transformers, estimation of time series models, and inference for dependence relationships

University of Illinois at Urbana-Champaign | MS in Statistics May 2022

- GPA: 3.95/4.00 | Awards: two years of guaranteed assistantships with full tuition waiver and stipend

Fordham University | BS in Mathematics with Minors in Computer Science and Economics May 2020

- GPA: 3.77/4.00 | Awards: *magna cum laude* | GRE: 170/170 Quantitative, 163/170 Verbal, 4.5/6.0 Writing

PROGRAMMING LANGUAGES AND TOOLS

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- Python expert: extensive experience with NumPy, pandas, Polars, scikit-learn, statsmodels, and PyTorch
 - R expert: extensive experience with dplyr, Rcpp, xts, zoo, caret, mgcv, glmnet, parallel, and ggplot2
 - SQL, Git, Bash expert: extensive experience with data management, version control, and Linux
 - C++ proficient: used for courses in Data Structures and Algorithms
 - q/kdb+ proficient: used during internship to work with high-frequency financial data

COURSEWORK

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- **Statistics:** Advanced Statistical Theory, Intermediate Statistics, Mathematical Statistics, Advanced Time Series Analysis, Regression Analysis, Computational Statistics
 - **Computer Science:** Algorithms, Data Structures, Theory of Computation, Operating Systems, Computer Architecture, Artificial Intelligence, Machine Learning, Data Mining for Listening to the Social Universe
 - **Math:** Stochastic Calculus, Measure-theoretic Probability, Functional Analysis, Measure Theory, Interacting Particle Systems, Geometric Flows, Differential Geometry, Lie Groupoids & Lie Algebroids, Topology, Abstract Algebra, Numerical Analysis, Numerical Linear Algebra, Real Analysis, Differential Equations, Linear Algebra, Mathematical Modeling

DOCTORAL RESEARCH

Foundation Models for Time Series, Invariant Prediction, and the Role of Statisticians | Aaditya Ramdas

- Developing a statistical framework for predicting nonstationary sequences using transformer-based foundation models

Invariant Prediction for Time-varying Sequences | Aaditya Ramdas

- Creating a statistical method that can be used to make machine learning models more robust to nonstationarity

Signature Methods for Large Language Models | Leif Weatherby, Tyler Shoemaker, and Cosma Shalizi

- Developing methods for analyzing sequences output by large language models using advanced stochastic process theory

Goodness-of-fit Testing and Confidence Sets through Random Features | Cosma Shalizi

- Creating a framework for uncertainty quantification and testing whether the law of a stochastic process is in a model class

Estimating Dynamic Models by Matching Random Features | Cosma Shalizi

- Developing a likelihood-free method for estimating the parameters of complicated models for time series

Conditional Independence Testing for Nonstationary Time Series | Michel F. C. Haddad and Aaditya Ramdas

- Creating a conditional independence test based on regression that is robust to nonstationarity and temporal dependence

GRADUATE RESEARCH ASSISTANTSHIPS

Carnegie Mellon University | [NSF Grant](#) | PI: Cosma Shalizi

May 2024-Present

- Developing theory, methods, and software in Python and R for estimating and inferring the parameters of dynamic models
- Using optimization algorithms to optimize highly non-convex objective functions to fit complicated models for time series
- Applied methods to time series models, state-space models, SDEs, ODEs, and dynamical systems with observational noise

University of Illinois at Urbana-Champaign | FORWARD Data Lab | Computer Science Department

Jan. 2021-May 2021

- Discovered patterns in cross-platform dynamics on Twitter, Facebook, and Reddit with Hawkes processes using Python

National Center for Supercomputing Applications | Innovative Software and Data Analysis Group

Sept. 2020-May 2022

- Built confidence bands for trends in concentrations and fluxes of chemicals to measure changes in water quality over time
- Used parallel computing to construct these confidence bands for 1000+ locations in the U.S. and made visualizations using R

INDUSTRY INTERNSHIPS

J.P. Morgan | Quantitative Research | Markets Summer Associate | Equity Derivatives and Macro

June 2023-Aug. 2023

- Worked with macro traders and quants on a hedging method for derivatives portfolios via multi-period optimization
- Collaborated with energy derivatives traders on improving the statistical techniques used in a new trading strategy

J.P. Morgan | Quantitative Research | Markets Summer Associate | Equity Derivatives

June 2022-Aug. 2022

- Developed method for adaptively selecting parameters of trade execution algorithms based on real-time data
- Built pipeline for processing market microstructure data using `q/kdb+` and making predictions using `scikit-learn` in Python
- Used Bayesian optimization and walk-forward cross-validation to tune the hyperparameters of machine learning models

RESEARCH INTERNSHIPS

MIT Lincoln Lab | Group 38

May 2021-July 2021

- Implemented optical tracking methods for tracking objects in outer space and ran simulations to compare approaches

Corteva Agriscience | Research & Development Division

June 2020-Aug. 2020

- Analyzed environmental changes based on time series of 50+ weather and soil features from 1,000+ sensor sites in the U.S.
- Used spatiotemporal Gaussian mixture models, dimensionality reduction, and clustering techniques with Python
- Built an interactive Dash application to visualize the statistical analysis of how environments changed over space and time

Fordham University | Robotics and Computer Vision Lab | Computer Science Department

Aug. 2019-March 2020

- Created a real-time tracking system for the Kihansi spray toad in collaboration with zoologists using deep learning in Python

TEACHING ASSISTANTSHIPS

- For the MS in Computational Finance program at CMU: Simulation Methods for Option Pricing, Financial Time Series, Financial Data Science I and II, and the Machine Learning Capstone Project (x2)
- For the MS in Data Science program at CMU: Time Series and Experimental Design
- For the BS in Statistics and Statistics/Machine Learning programs at CMU: Advanced Data Analysis

PROFESSIONAL SERVICE

- 2026: reviewer of one paper for the Journal of the Royal Statistical Society, Series B (Statistical Methodology)
- 2024-2025: chair of the Statistics PhD student committee at Carnegie Mellon University focused on organizing career events

TALKS AND POSTERS

- 2026: in-person talk on conditional independence testing at the International Workshop in Sequential Methodologies at American University
- 2025: in-person talk on simulation-based inference by matching random features at the Statistical Methods for the Physical Sciences (STAMPS) Research Center's local meeting at Carnegie Mellon University
- 2025: poster on simulation-based inference by matching random features at the Workshop on Neural Simulation-based Inference at Carnegie Mellon University

- 2025: online talk on variable selection for large-scale forecasting for nonstationary time series to researchers at Amazon
- 2025: online talk on conditional independence testing at the Virtual Time Series Seminar's Workshop for Junior Researchers
- 2024: poster on conditional independence testing at the NBER-NSF Time Series Conference at University of Pennsylvania
- 2024: in-person talk on regression for time series at Statistics and Machine Learning Group at Carnegie Mellon University